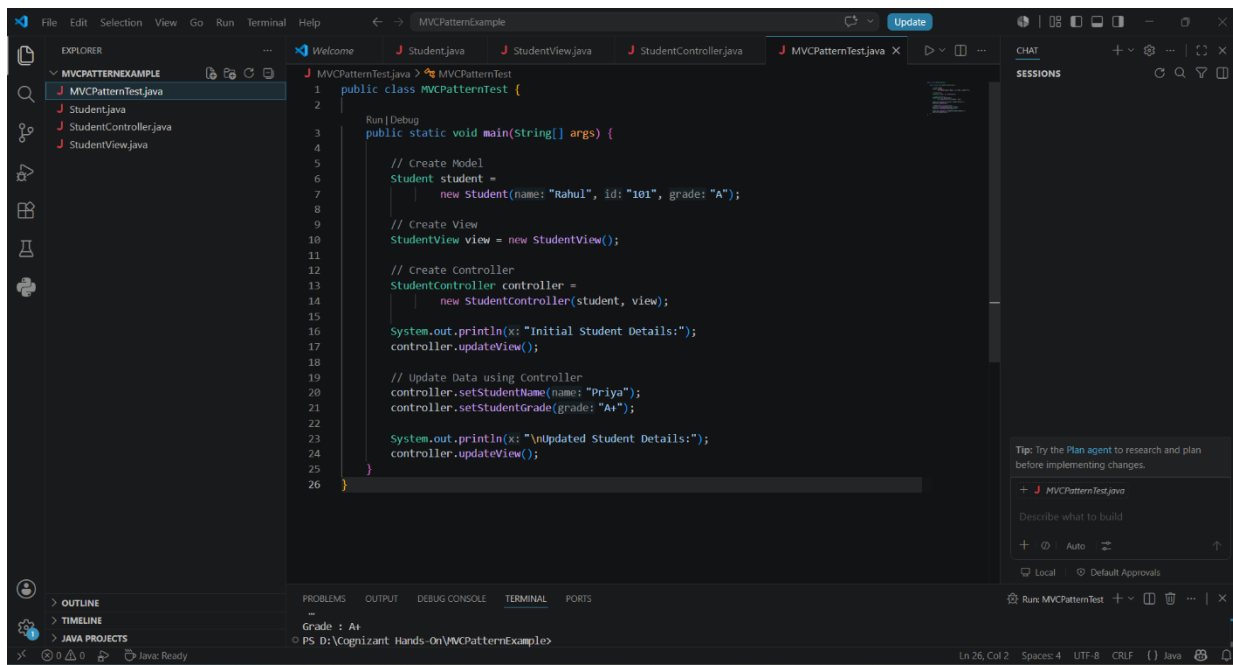


Hands-On

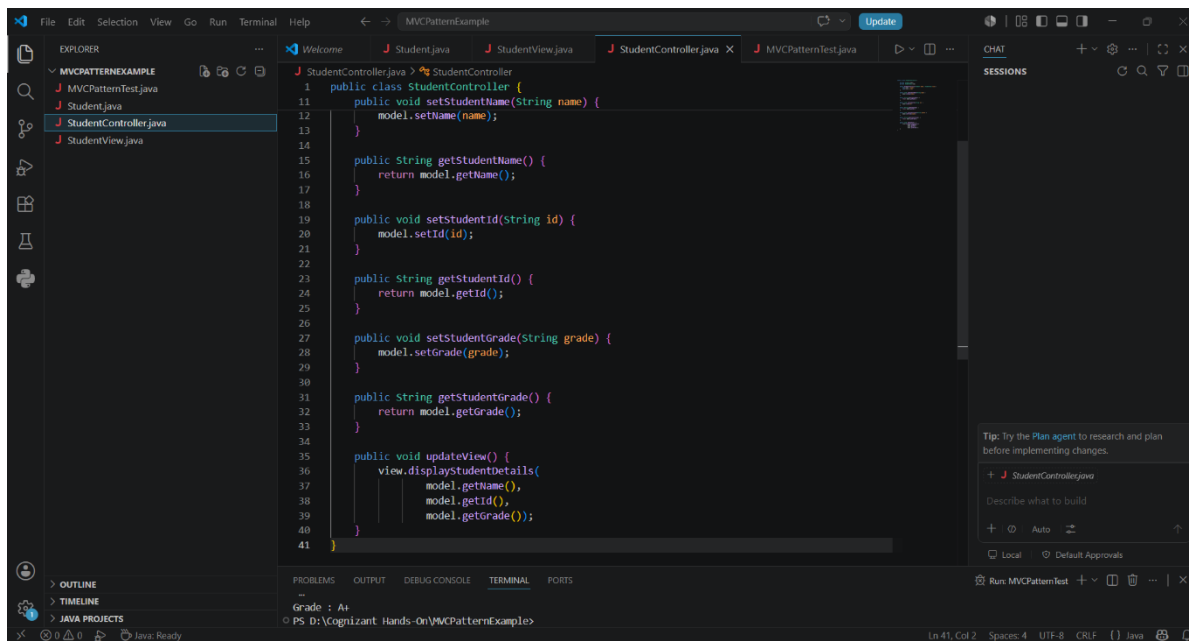
Exercise 10: Implementing the MVC Pattern

In this module, I have implemented, executed and tested Model-View-Controller (MVC) Pattern in Java with the help of a project called MVCPatternExample. I have developed a Student class to store the student data including name, student id, grade etc., a StudentView class to show the student data and a StudentController class to handle the communication between the model and the view. The controller was responsible for getting, updating and passing data from the Student class to the StudentView class and vice versa. Once the implementation was done, I tested the application by running it and creating and updating student records via the controller, while showing the updated information in the view. This practical exercise was beneficial to me in understanding the MVC Pattern which helps to separate the data, business logic, and presentation layers, making the code easier to organize, maintain, scalable, and the structure of software applications more organized.

Coding:



```
1 public class MVCPatternTest {
2
3     Run | Debug
4     public static void main(String[] args) {
5
6         // Create Model
7         Student student =
8             new Student(name: "Rahul", id: "101", grade: "A");
9
10        // Create View
11        StudentView view = new StudentView();
12
13        // Create Controller
14        StudentController controller =
15            new StudentController(student, view);
16
17        System.out.println(x: "Initial Student Details:");
18        controller.updateView();
19
20        // Update Data using Controller
21        controller.setStudentName(name: "Priya");
22        controller.setStudentGrade(grade: "A+");
23
24        System.out.println(x: "\nUpdated Student Details:");
25        controller.updateView();
26    }
```



Output:

